

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tool

- Diesel exhaust fluid (DEF) extension kit, Part Number 5299873
- Graduated beaker, Part Number 4919139, or equivalent.
- Cummins® electronic service tool or equivalent

Additional Service Items

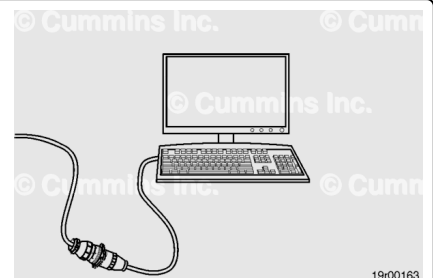
- No additional service items required.

General Information

The purpose of this test is to run the aftertreatment DEF dosing system through a dosing cycle and check the:

- Aftertreatment DEF dosing valve spray characteristics
- Amount of DEF delivered in a specified time.

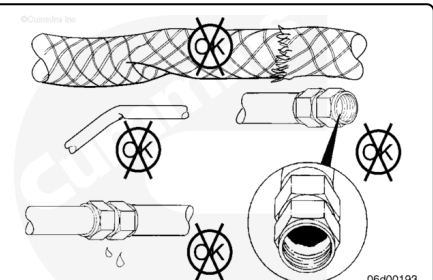
The test can be accessed through the recommended Cummins® electronic service tool or equivalent under engine control module (ECM) Diagnostic Tests. Follow the on-screen instructions to perform the test.



Initial Check

Check for any leaks, blockages, or restrictions in the DEF line between the aftertreatment DEF dosing unit and the aftertreatment DEF dosing valve.

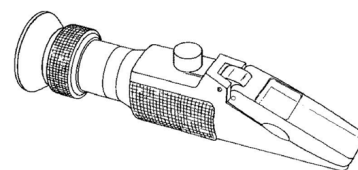
Repair as necessary. See equipment manufacturer service information.



Check that there is an adequate amount of DEF in the DEF tank prior to starting this test.

Check the concentration and quality of the DEF. Refer to Procedure 011-056 in Section 11.

(/qs3/pubsys2/xml/en/procedures/493/493-011-056.html)



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Allow for DEF to recirculate and depressurize after engine shut down and prior to DEF system servicing. This can take up to 15 minutes.

There is **always** a chance of residual pressure being present. Open fittings slowly to allow any pressure to bleed off before removing any connections.

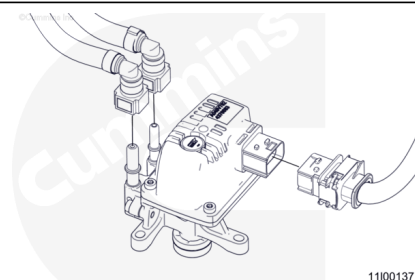
Note : Low battery voltage can cause the dosing volume to be low. Check the batteries. See equipment manufacturer service information.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the aftertreatment DEF dosing valve from the decomposition tube. Refer to Procedure 011-059 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-059.html)

Setup

⚠ CAUTION ⚠

Take care when handling and/or disconnecting the DEF line from the aftertreatment DEF dosing valve. The intake of the aftertreatment DEF dosing valve is plastic and can be easily damaged.



⚠ CAUTION ⚠

Do not connect a 12 VDC or 24 VDC supply to the aftertreatment DEF dosing valve as this will cause permanent damage.

Connect the electrical and DEF lines. Refer to Procedure 011-059 in Section 11.

(/qs3/pubsys2/xml/en/procedures/493/493-011-059.html)

Note : There may not be enough length for the existing electrical and/or DEF line to allow the aftertreatment DEF dosing valve to be placed in the container. Use service tool kit, Part Number 5299873, or equivalent, which contains:

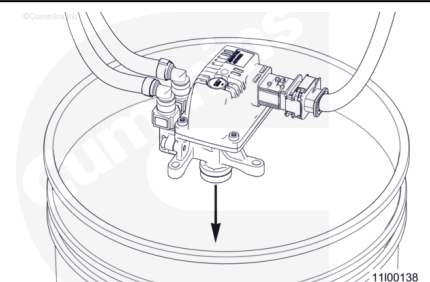
- Wiring harness extension, Part Number 5299874. The wiring harness extension can be placed in between the original equipment manufacturer (OEM) wiring harness and the DEF dosing valve.
- DEF extension line, Part Number 5299875. This extension line can be placed in between the DEF pressure line and the DEF dosing valve.
- DEF extension line, Part Number 5299912. This extension line can be placed in between the DEF return line and the DEF dosing valve.

Note : It may be easier to capture the DEF in a clean container and transfer the DEF to the measuring device for the final measurement.

Obtain a clear plastic container (large enough to hold the aftertreatment DEF dosing valve) and a graduated beaker, Part Number 4919139, or equivalent. A measuring cup that is marked in milliliters (ml) or ounces (oz) can also be used.

The measuring device **must** be capable of measuring between 0 ml [0.0 oz] and 500 ml [17.0 oz] in 5 ml [0.34 oz] increments

Place the aftertreatment DEF dosing valve into the container.



⚠ WARNING ⚠

Diesel exhaust fluid (DEF) contains urea. Do not get the substance in your eyes. In case of contact, immediately flush eyes with large amounts of water for a minimum of 15 minutes. Do not swallow. In the event the DEF is ingested, contact a physician immediately. Reference the materials safety data sheet (MSDS) for additional information.



During the test, it will be necessary to monitor or log DEF pressure in the electronic service tool.

Note : When the test is started, the dosing system will first prime. During this process, the aftertreatment DEF dosing valve will open intermittently to purge air from the system. In doing so, some DEF will be sprayed from the tip. This is a normal operating characteristic.

When the test is being performed, the aftertreatment DEF dosing valve will spray a very fine mist of DEF. To prevent fine mist from escaping into the air and to make sure of an accurate measurement, place a clean shop towel or cover over the valve and container.

When the test begins, briefly monitor the spray pattern of the DEF exiting the aftertreatment DEF dosing valve. Check for:

- Signs of larger drops and/or dripping DEF from the tip
- Spray pattern that is **not** symmetrical (sprays more to one side).

If either of these symptoms is noted, stop the test and inspect the aftertreatment DEF dosing valve tip. Refer to Procedure 011-059 in Section 11.

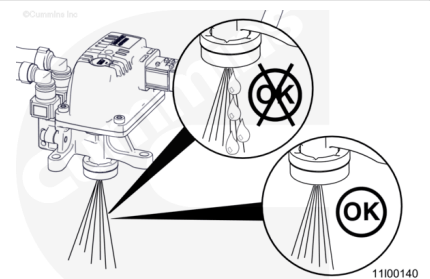
(/qs3/pubsys2/xml/en/procedures/493/493-011-059.html)

After cleaning, run the test again and monitor the spray pattern. If the problem persists, replace the aftertreatment DEF dosing valve. Refer to Procedure 011-059 in Section 11.

(/qs3/pubsys2/xml/en/procedures/493/493-011-059.html)

Note : If a leak at the DEF dosing valve tip is suspected, then the Aftertreatment Diesel Exhaust Fluid System Leak Test may be used to confirm the symptom. Refer to Procedure 011-080 in Section 11.

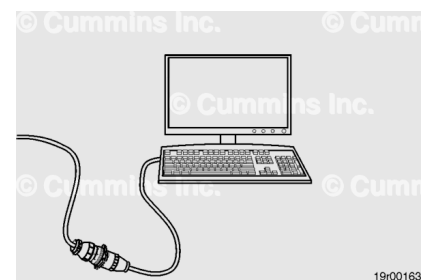
(/qs3/pubsys2/xml/en/procedures/493/493-011-080.html)



Flow Test

Note : Prior to performing this test, if not already directed by a fault code troubleshooting tree, view and troubleshoot any fault codes with the electronic service tool.

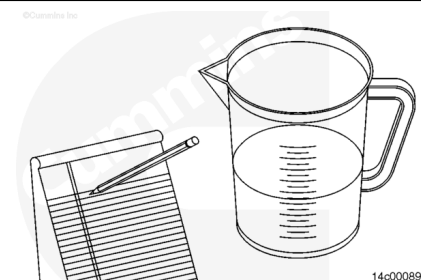
- Turn the keyswitch ON.
- Connect the electronic service tool.
- Locate the Diesel Exhaust Fluid Doser Pump Override Test under ECM Diagnostic Tests.
- Follow the on-screen instructions to perform the test.



The electronic service tool will start the test and will inject the DEF. The electronic service tool will automatically stop the injector at the end of the test. If the test needs to be stopped before finishing, click the Stop button.

After the test is complete, measure the amount of DEF sprayed into the container. Pour the DEF into the graduated beaker, Part Number 4919139, or equivalent.

Perform the test three times. The amount of DEF measured for each test **must** be within specification.



Aftertreatment DEF Dosing Valve Volume Specifications

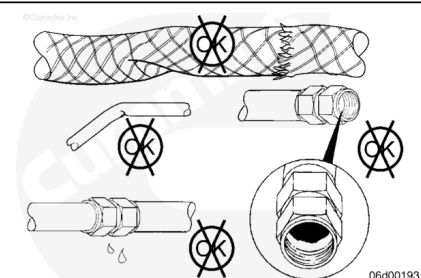
ml		fl-oz
255	MIN	8.6
345	MAX	11.7

Note : Do not pour the DEF back into the DEF tank. Dispose of the DEF in accordance with local environmental regulations.

Note : Inspect the collected sample of DEF for signs of debris or contamination.

Note : Low battery voltage can cause the dosing volume to be low. Check the batteries. See equipment manufacturer service information. View and troubleshoot any fault codes with the electronic service tool.

If DEF pressure drops below 700 kPa [101.5 psi] or is erratic during the test, the following **must** be checked regardless of whether or **not** the amount of DEF collected was within specification. It is normal for pressure to drop initially when the DEF dosing valve opens, but should recover in a couple seconds.



- Air in the DEF. Refer to Procedure 011-075 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-075.html>)

- DEF contaminated with debris.
- Large leak in the DEF pressure line.
- Inspect the DEF dosing unit intake fitting for debris. Refer to Procedure 011-058 in Section 11.
(/qs3/pubsys2/xml/en/procedures/493/493-011-058.html)
- If debris is found in the intake fitting, replace the fitting and perform the DEF dosing unit manual purge. Refer to Procedure 011-143 in Section 11.
(/qs3/pubsys2/xml/en/procedures/493/493-011-143.html) Also, drain and flush the DEF tank and intake line. See equipment manufacturer service information.

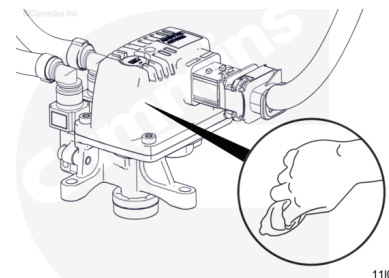
Use the following as a guide if the amount of DEF is **not** within specification for one or more of the three tests and DEF pressure is stable for the duration of the tests:

- Clean the DEF dosing valve if one or more of the test results are below specification.
- Replace the DEF dosing valve if one or more of the test results are above specification.
- Replace the DEF dosing valve if the test results vary with both below-specification and above-specification conditions. This could be an erratic valve with intermittent malfunction and **must** be replaced. Refer to Procedure 011-059 in Section 11.
(/qs3/pubsys2/xml/en/procedures/493/493-011-059.html)

Note : The aftertreatment DEF dosing valve may have been plugged by debris. Inspect the aftertreatment DEF dosing unit filter for signs of contamination and debris prior to installing the new aftertreatment DEF dosing valve. Refer to Procedure 011-060 in Section 11.
(/qs3/pubsys2/xml/en/procedures/493/493-011-060.html)

⚠ CAUTION ⚠

Do not submerge the aftertreatment DEF dosing valve in solvent or water. Damage to the aftertreatment DEF dosing valve will result.



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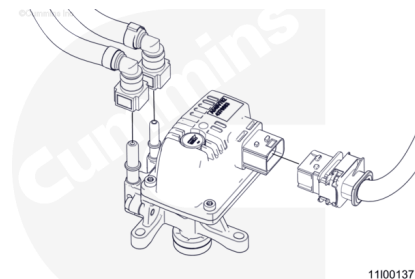
After the test is complete, a light coating of DEF will cover the aftertreatment DEF dosing valve, DEF line, and electrical line.

Prior to disconnecting anything, use a clean shop towel soaked in warm water to wipe the coating of DEF from the components.

Note : Clean any service tools used for this test with warm, distilled water before storage.

Disconnect the electrical and DEF lines. Refer to Procedure 011-059 in Section 11.

(/qs3/pubsys2/xml/en/procedures/493/493-011-059.html) If used, remove the service tool extensions.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the aftertreatment DEF dosing valve. Refer to Procedure 011-059 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-059.html)
- Connect the batteries. See equipment manufacturer service information.